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Math 3108 – Linear Algebra
Exam 3 – Part B
Prof. Runnion
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Directions:

- 1) This packet is a continuation of Part A. All directions on Part A still apply.
- 2) This packet has three sheets of paper, including this cover page. Do NOT remove the staple or remove any sheet from this packet.

	4	5	Part B
Points Earned	23	12	35
Maximum Points	23	14	37

**DO NOT OPEN THIS EXAM
UNTIL TOLD TO DO SO**

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Open Answer. Show ALL your work!

11. Consider the matrix $A = \begin{bmatrix} 1 & -6 & 9 & 0 & -2 \\ 0 & 1 & 2 & -4 & 5 \\ 0 & 0 & 0 & 5 & 1 \\ 0 & 0 & 0 & 10 & 2 \end{bmatrix}$

$$\xrightarrow{\frac{1}{10}R_4} \begin{bmatrix} 1 & -6 & 9 & 0 & -2 \\ 0 & 1 & 2 & -4 & 5 \\ 0 & 0 & 0 & 5 & 1 \\ 0 & 0 & 0 & 1 & \frac{1}{5} \end{bmatrix} \xrightarrow{-5R_4 + R_3 \rightarrow R_3}$$

$$\begin{bmatrix} 1 & -6 & 9 & 0 & -2 \\ 0 & 1 & 2 & -4 & 5 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & \frac{1}{5} \end{bmatrix}$$

a) (3 points) The rank of A is 3.

b) (3 points) The dimension of $\text{Nul } A$ is 2.

c) (3 points) Find a basis for the row space of A .

$$\{(1, -6, 9, 0, -2); (0, 1, 2, -4, 5); (0, 0, 0, 1, \frac{1}{5})\}$$

12. (10 points) Given the matrix $A = \begin{bmatrix} 5 & -2 & 6 & -1 \\ 0 & 3 & h & 0 \\ 0 & 0 & 5 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, find the value for h such that the

eigenspace corresponding to $\lambda = 5$ is two-dimensional.

$h = 5$:

$$\begin{bmatrix} 0 & -2 & 6 & -1 & | & 0 \\ 0 & -2 & h & 0 & | & 0 \\ 0 & 0 & 0 & 4 & | & 0 \\ 0 & 0 & 0 & -4 & | & 0 \end{bmatrix}$$

Both need
To be Pivots

$\hookrightarrow$ Only Want 2 Pivots
So you want 3 Free
Variables

$$\therefore \boxed{h = 6}$$

13. (4 points) Find a unit vector $\mathbf{u}$ in the direction of $\mathbf{v} = \begin{bmatrix} -1 \\ 2 \\ 0 \\ -2 \end{bmatrix}$

$$\frac{\mathbf{v}}{\|\mathbf{v}\|} \Rightarrow \frac{1}{3} \begin{bmatrix} -1 \\ 2 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} -1/3 \\ 2/3 \\ 0 \\ -2/3 \end{bmatrix}$$

$$\|\mathbf{v}\| = \sqrt{1^2 + 2^2 + 0^2 + 2^2} = \sqrt{1 + 4 + 4} = 3$$

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14. (14 points) The vectors $x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, $x_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, and $x_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$ form a basis for a subspace W of $\mathbb{R}^4$. Use the Gram-Schmidt process to produce an orthogonal basis $\{v_1, v_2, v_3\}$ for W where $x_1 = v_1$.

$$\bar{x}_1 = \bar{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{aligned} \bar{v}_2 &= x_2 - \frac{x_2 \cdot \bar{v}_1}{\bar{v}_1 \cdot \bar{v}_1} \bar{v}_1 = x_2 - \frac{0(1) + 1(1) + 1(1) + 1(1)}{1^2 + 1^2 + 1^2 + 1^2} \bar{v}_1 \\ &= x_2 - \frac{3}{4} \bar{v}_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{3}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -3/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \bar{v}_3 &= x_3 - \frac{x_3 \cdot \bar{v}_1}{\bar{v}_1 \cdot \bar{v}_1} \bar{v}_1 - \frac{x_3 \cdot \bar{v}_2}{\bar{v}_2 \cdot \bar{v}_2} \bar{v}_2 \\ &= \bar{x}_3 - \frac{0(1) + 0(1) + 1(1) + 1(1)}{4} \bar{v}_1 - \frac{0(-3/4) + 0(1/4) + 1(1/4) + 1(1/4)}{(-3/4)^2 + (1/4)^2 + (1/4)^2 + (1/4)^2} \bar{v}_2 \\ &= \bar{x}_3 - \frac{1}{2} \bar{v}_1 - \frac{2}{3} \bar{v}_2 \end{aligned}$$

$\frac{1}{2} \cdot \frac{4}{3} = \frac{4}{6} = \frac{2}{3}$

$$\begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} -1/2 \\ -1/2 \\ 1/2 \\ 1/2 \end{bmatrix} - \frac{2}{3} \begin{bmatrix} -3/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}$$

Basis: $\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -3/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}, \begin{bmatrix} -1/2 \\ -1/2 \\ 1/2 \\ 1/2 \end{bmatrix} \right\}$